



Name: _____

2023

**YEAR 11
ASSESSMENT TASK 3**

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black or blue pen
- Diagrams are not to scale
- Calculators may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
64**

Section I – 10 marks (page 2-4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 54 marks (pages 5–20)

- Attempt Questions 11–13
- Allow about 1 hour 45 minutes for this section

BLANK PAGE

Section I: Multiple Choice Questions**10 marks***Circle the correct answer.*

1. What is the derivative of $\frac{2}{x^3}$?

- A. $-\frac{3}{x^4}$
- B. $\frac{6}{x^4}$
- C. $\frac{3}{x^4}$
- D. $-\frac{6}{x^4}$

2. What is the value of $\tan x$ given that $\sin x = \frac{2}{5}$ for $\frac{\pi}{2} \leq x \leq \pi$

- A. $-\frac{\sqrt{21}}{2}$
- B. $-\frac{5}{\sqrt{21}}$
- C. $\frac{2}{\sqrt{21}}$
- D. $-\frac{2}{\sqrt{21}}$

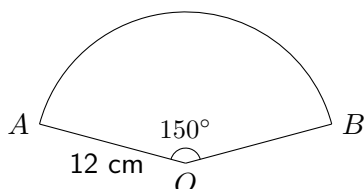
3. Which of the following is the equation of the line that passes through $(2, -1)$ and $(5, 3)$?

- A. $4x + 3y - 11 = 0$
- B. $3x - 4y + 11 = 0$
- C. $4x - 3y - 11 = 0$
- D. $3x - 4y - 11 = 0$

4. Which of the following is equivalent to $7x^2 - 56x + 105$?

- A. $(x - 8)(x + 7)$
- B. $7(x - 3)(x - 5)$
- C. $(x - 3)(x - 5)$
- D. $7(x - 8)(x + 7)$

5. What is the length of arc AB in the diagram below?



- A. 10π cm
- B. 5 cm
- C. 5π cm
- D. 10 cm

6. The domain for the function $f(x) = \sqrt{2-x}$ is:

- A. $[0, \infty)$
- B. $(-\infty, 2]$
- C. $[2, \infty)$
- D. $(-\infty, \infty)$

7. Solve $\tan x = 1$ for $0 \leq x \leq 2\pi$.

- A. $x = \frac{\pi}{4}, \frac{5\pi}{4}$
- B. $x = \frac{\pi}{4}$
- C. $x = \frac{\pi}{4}, \frac{3\pi}{4}$
- D. $x = \frac{\pi}{4}, \frac{7\pi}{4}$

8. Megan tries to differentiate the function $y = \frac{3x}{(10-x)^2}$ with respect to x .

Her working out is shown below.

$$\begin{aligned} \frac{dy}{dx} &= \frac{3(10-x)^2 + 3x \times 2(10-x)}{(10-x)^4} && \text{Line (1)} \\ &= \frac{3(10-x)^2 - 6x(10-x)}{(10-x)^4} && \text{Line (2)} \\ &= \frac{(10-x)[3(10-x) + 6x]}{(10-x)^4} && \text{Line (3)} \\ &= \frac{30 + 3x}{(10-x)^3} && \text{Line (4)} \\ &= \frac{10 + x}{(10-x)^3} && \text{Line (5)} \end{aligned}$$

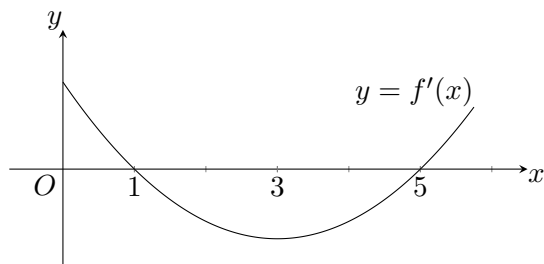
Where did Megan make mistakes?

- A. Line 1 & Line 5
- B. Line 3 & Line 4
- C. Line 1 & Line 3
- D. Line 3 & Line 5

9. The height of a ball thrown in the air is given by the formula $h = 10t - 3t^2$ in metres, where t is measured in seconds. When the ball is thrown, what is the average rate of change of the height in the first second?

- A. 4 metres per second
- B. 7 metres
- C. $10 - 6t$
- D. 7 metres per second

10. The diagram below shows the graph of the gradient function of a curve.



For what value(s) of x does $f(x)$ have a local minimum?

- A. $x = 1$
- B. $x = 3$
- C. $x = 1, x = 5$
- D. $x = 5$

End of Section I

Section II**54 marks***Answer the questions in the spaces provided.**Show all necessary working.***Question 11** (1 marks)**1**Show that $f(x) = x^3 + 3x$ is an odd function.

Question 12 (2 marks)**2**What is the centre and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$?

Question 13 (2 marks)**2**Express $\frac{4}{4 - \sqrt{13}}$ in the form $a + \sqrt{b}$, where a and b are integers.

Question 14 (7 marks)Differentiate the following with respect to x .

(a) $y = 3x^5 - 2x^3 + 7$

1

(b) $y = (x^2 - 2)^3(x + 3)^2$

2

(c) $y = \sqrt[4]{4x + 3}$

2

(d) $y = \frac{2x}{5 - x}$

2

Question 15 (3 marks)**3**

Find the equation of the normal to the curve $y = 2x^2 - 5x + 1$ at the point $(2, -1)$

Question 16 (2 marks)**2**

Differentiate $f(x) = 2x + 3$ from first principles to show that $f'(x) = 2$.

Hint: Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

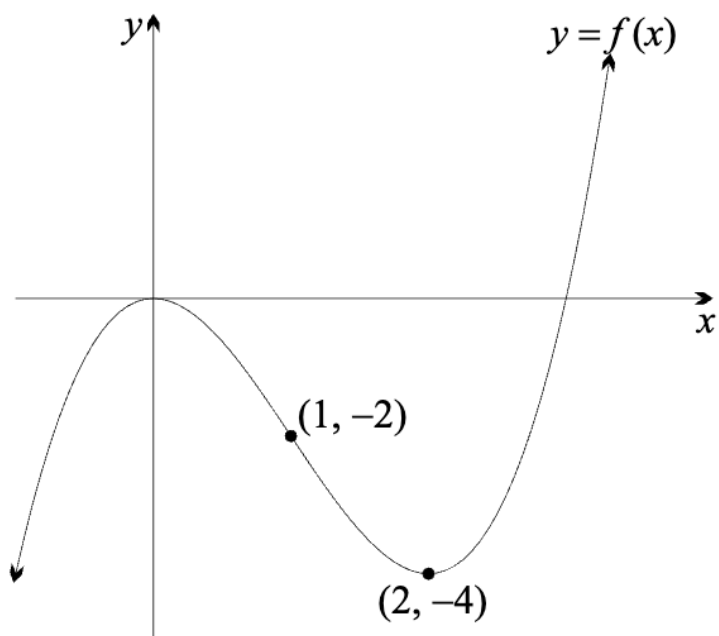
Question 17 (2 marks)**2**

Let $f(x) = 2x$ and $g(x) = \frac{x}{x+1}$. Solve $f(g(x)) = 3$.

Question 18 (2 marks)**2**

The function $y = f(x)$ is graphed in the diagram below. The points $(2, -4)$ and the origin are stationary points and the point $(1, -2)$ is a point of inflexion.

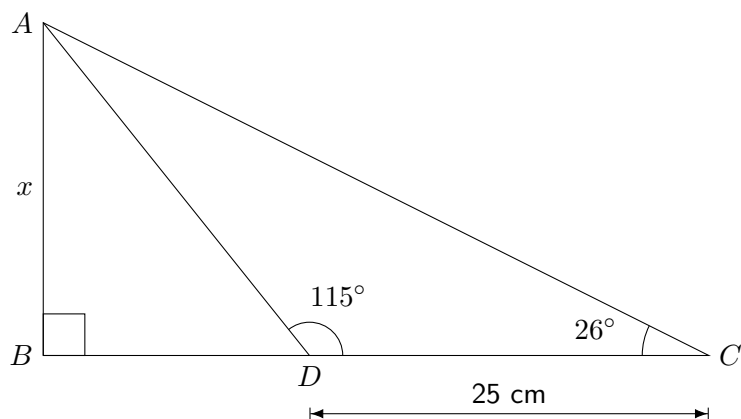
Sketch the graph of the curve $y = f'(x)$ on the same cartesian plane, showing the shape of the curve and any x intercepts.



Question 19 (3 marks)

3

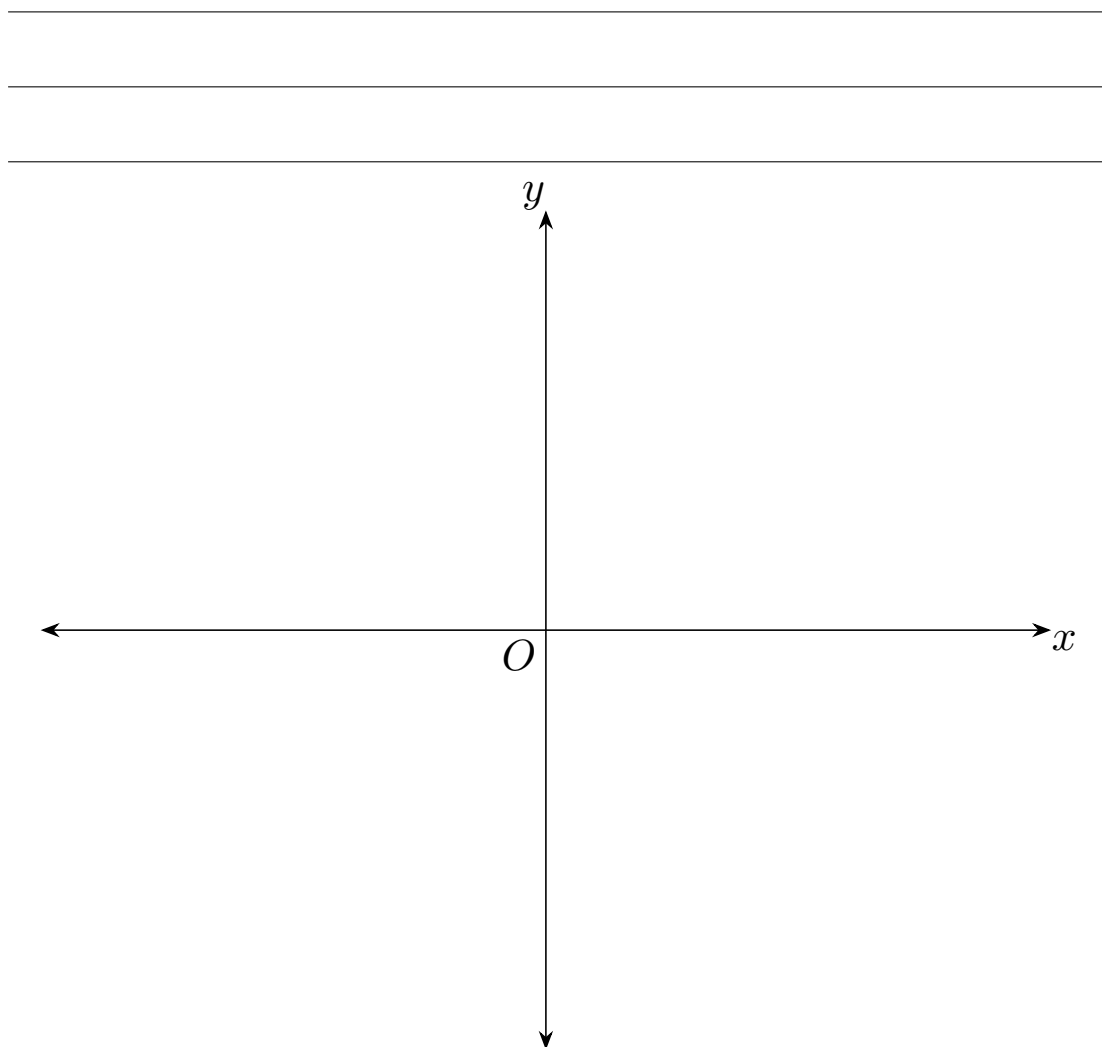
Find the value of x , correct to 2 decimal places.

[illegible]

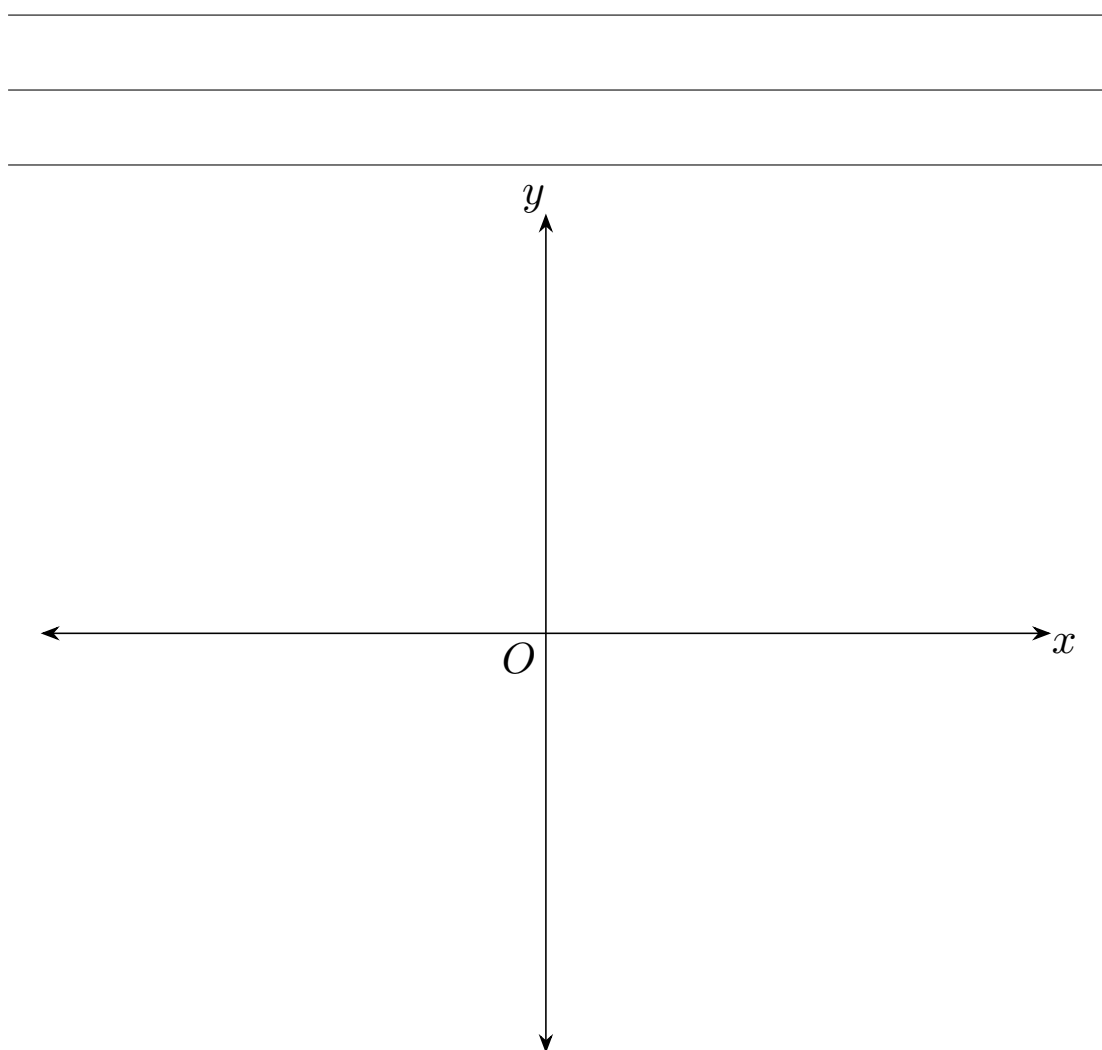
Question 20 (4 marks)

Sketch the graphs of the following, showing any asymptotes and any intercepts with the axes.

(a) $y = \frac{2}{2x - 5}$

2

(b) $y = |4x + 5|$

2

Question 22 (3 marks)

Consider the function $x^4 - 12x^2 + 32 = 0$.

(a) Solve for x by first letting $u = x^2$.

2

(b) If $x = \sin \theta$, explain why there are no solutions for θ .

1

Question 23 (8 marks)

Consider the curve $f(x) = x^3 - 6x^2 + 9x - 4$.

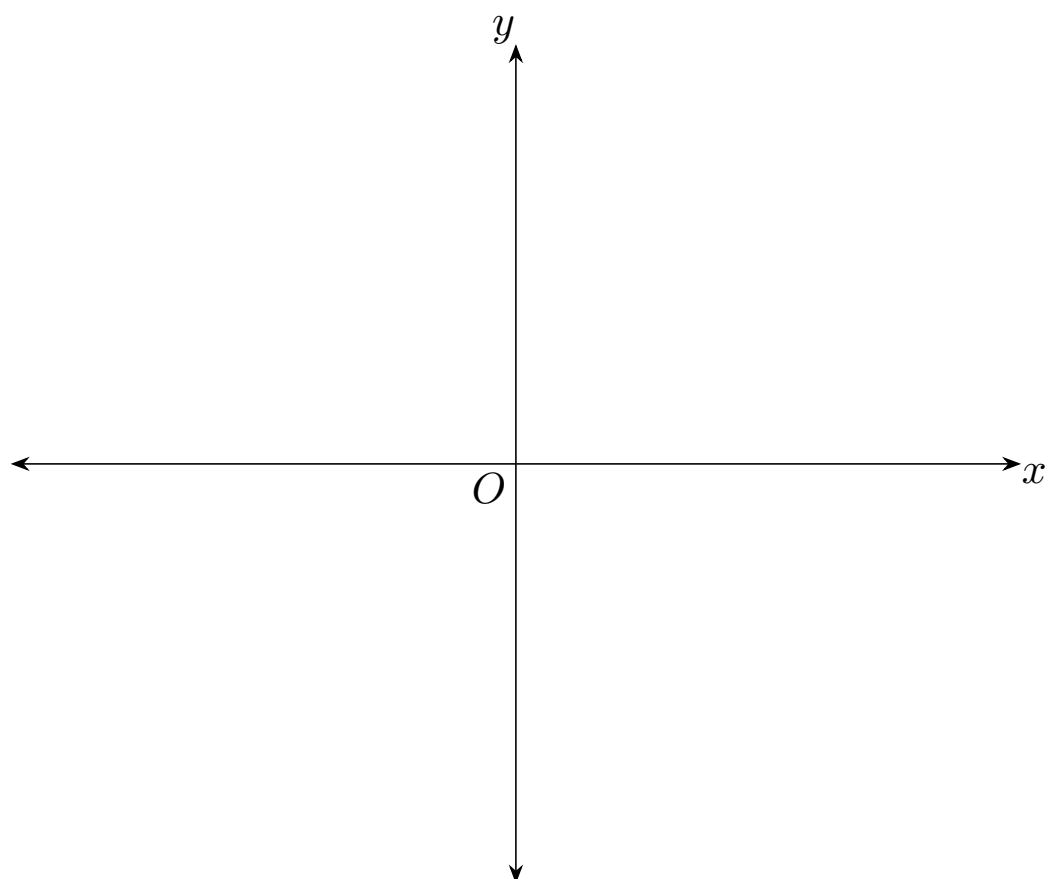
- (a) Find any stationary points and determine the nature.

3

- (b) Find the co-ordinates of the point of inflexion.

2

- (c) Hence, sketch the curve, showing all turning points, the y -intercept and the point of inflexion. **3**

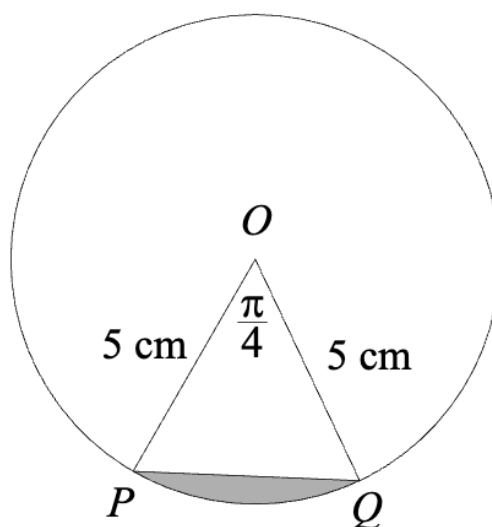


Question 24 (4 marks)

4

Solve $2 \cos^2 x + 3 \sin x = 0$ for $0 \leq x \leq 2\pi$.

[illegible]

Question 25 (3 marks)

The diagram above shows a circle with centre O , radius 5 cm and $\angle POQ = \frac{\pi}{4}$. The points P and Q lie on the circumference of the circle.

- (a) Find the exact area of the sector OPQ .

1

- (b) Find the exact area of $\triangle OPQ$

1

- (c) Hence find the exact area of the minor shaded segment.

1

Question 26 (5 marks)

Farmer Anneka wants to make a rectangular paddock with an area of 4000m^2 .

However, fencing costs are sky-high so she wants the paddock to have a minimum perimeter.

- (a) Show that the perimeter is given by the equation $P = 2x + \frac{8000}{x}$.

2

- (b) Find the dimensions of the rectangle that will give the minimum perimeter.
Give your answer correct to 1 decimal place.

3

[illegible]

End of Task



Name: _____

2023

YEAR 11
ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black or blue pen
- Diagrams are not to scale
- Calculators may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
64

Section I – 10 marks (page 2-4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 54 marks (pages 5–20)

- Attempt Questions 11–13
- Allow about 1 hour 45 minutes for this section

BLANK PAGE

Section I: Multiple Choice Questions**10 marks***Circle the correct answer.*

1. What is the derivative of $\frac{2}{x^3}$?

- A. $-\frac{3}{x^4}$
 B. $\frac{6}{x^4}$
 C. $\frac{3}{x^4}$
 D. $-\frac{6}{x^4}$

$$y = 2x^{-3}$$

$$y' = -6x^{-4}$$

$$= -\frac{6}{x^4}$$

2. What is the value of $\tan x$ given that $\sin x = \frac{2}{5}$ for $\frac{\pi}{2} \leq x \leq \pi$?

- A. $-\frac{\sqrt{21}}{2}$
 B. $-\frac{5}{\sqrt{21}}$
 C. $\frac{2}{\sqrt{21}}$
 D. $-\frac{2}{\sqrt{21}}$



3. Which of the following is the equation of the line that passes through $(2, -1)$ and $(5, 3)$?

- A. $4x + 3y - 11 = 0$
 B. $3x - 4y + 11 = 0$
 C. $4x - 3y - 11 = 0$
 D. $3x - 4y - 11 = 0$

$$y_1 = -1, x_1 = 2; y_2 = 3, x_2 = 5$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - (-1)}{5 - 2} = \frac{4}{3}$$

$$y + 1 = \frac{4}{3}(x - 2)$$

$$3y + 3 = 4x - 8$$

$$0 = 4x - 3y - 11$$

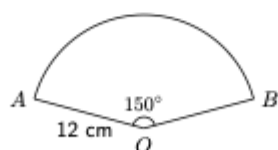
4. Which of the following is equivalent to $7x^2 - 56x + 105$?

- A. $(x - 8)(x + 7)$
 B. $7(x - 3)(x - 5)$
 C. $(x - 3)(x - 5)$
 D. $7(x - 8)(x + 7)$

$$7(x^2 - 8x + 15)$$

$$= 7(x - 3)(x - 5)$$

5. What is the length of arc AB in the diagram below?



- A. 10π cm
 B. 5 cm
 C. 5π cm
 D. 10 cm

$$\frac{150^\circ}{180} = \frac{5\pi}{6}$$

$$l = r\theta$$

$$l = 12 \times \frac{5\pi}{6}$$

6. The domain for the function $f(x) = \sqrt{2-x}$ is:

A. $[0, \infty)$
 B. $(-\infty, 2]$
 C. $[2, \infty)$
 D. $(-\infty, \infty)$

$$\begin{aligned} 2-x &\geq 0 \\ -x &\geq -2 \\ x &\leq 2 \end{aligned}$$

7. Solve $\tan x = 1$ for $0 \leq x \leq 2\pi$.

A. $x = \frac{\pi}{4}, \frac{5\pi}{4}$
 B. $x = \frac{\pi}{4}$
 C. $x = \frac{\pi}{4}, \frac{3\pi}{4}$
 D. $x = \frac{\pi}{4}, \frac{7\pi}{4}$

8. Megan tries to differentiate the function $y = \frac{3x}{(10-x)^2}$ with respect to x .

Her working out is shown below.

ambiguous

$$\begin{aligned} \frac{dy}{dx} &= \frac{3(10-x)^2 + 3x \times 2(10-x)}{(10-x)^4} && \text{Line (1)} \\ &= \frac{3(10-x)^2 - 6x(10-x)}{(10-x)^4} && \text{Line (2)} \\ &= \frac{(10-x)[3(10-x) + 6x]}{(10-x)^4} && \text{Line (3)} \\ &= \frac{30 + 3x}{(10-x)^3} && \text{Line (4)} \\ &= \frac{10+x}{(10-x)^3} && \text{Line (5)} \end{aligned}$$

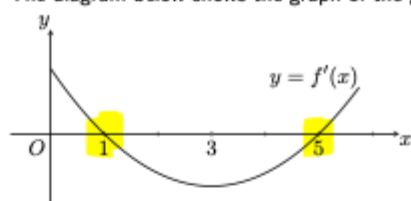
Where did Megan make mistakes?

A. Line 1 & Line 5
 B. Line 3 & Line 4
 C. Line 1 & Line 3
 D. Line 3 & Line 5

9. The height of a ball thrown in the air is given by the formula $h = 10t - 3t^2$ in metres, where t is measured in seconds. When the ball is thrown, what is the average rate of change of the height in the first second?

A. 4 metres per second
 B. 7 metres
 C. $10 - 6t$
 D. 7 metres per second

10. The diagram below shows the graph of the gradient function of a curve.



For what value(s) of x does $f(x)$ have a local minimum?

- A. $x = 1$
- B. $x = 3$
- C. $x = 1, x = 5$
- ☒ D. $x = 5$

End of Section I

Section II**54 marks**

Answer the questions in the spaces provided.
Show all necessary working.

Question 11 (1 marks)**1**

Show that $f(x) = x^3 + 3x$ is an odd function.

$$\begin{aligned} f(-x) &= (-x)^3 + 3(-x) \\ &= -x^3 - 3x \\ &= -(x^3 + 3x) \\ \therefore f(-x) &= -f(x) \\ \therefore \text{odd} \end{aligned}$$

① Mark for correct answer

Question 12 (2 marks)**2**

What is the centre and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$?

$$\begin{aligned} x^2 - 6x + y^2 + 4y &= 12 \\ x^2 - 6x + 9 + y^2 + 4y + 4 &= 12 + 9 + 4 \\ (x-3)^2 + (y+2)^2 &= 25 \\ \therefore \text{centre: } (3, -2) \quad r &= 5 \end{aligned}$$

① Mark for correctly completing the square.

① Mark for correct centre and radius

Question 13 (2 marks)**2**

Rationalise the denominator of $\frac{\sqrt{2}+4}{2\sqrt{2}-1}$. Write your answer in the form $a + b\sqrt{2}$, where a and b are rational.

$$\begin{aligned} \frac{\sqrt{2}+4}{2\sqrt{2}-1} &\times \frac{2\sqrt{2}+1}{2\sqrt{2}+1} \\ \frac{(\sqrt{2}+4)(2\sqrt{2}+1)}{7} &= \frac{4+5\sqrt{2}+8\sqrt{2}+4}{7} \\ &= \frac{8+13\sqrt{2}}{7} \\ \therefore a &= \frac{8}{7} \quad b = \frac{13}{7} \end{aligned}$$

① Mark for multiplying with conjugate

① Mark for correct solution.

Question 14 (7 marks)Differentiate the following with respect to x .

(a) $y = 3x^5 - 2x^3 + 7$

1

$$y' = 15x^4 - 6x^2$$

① Mark for correct answer

(b) $y = (x^2 - 2)^3(x + 3)^2$

2

$$u = (x^2 - 2)^3 \quad v = (x + 3)^2$$

$$u' = 6x(x^2 - 2)^2 \quad v' = 2(x + 3)$$

① mark for correct chain rule

$$y' = 6x(x^2 - 2)^2(x + 3)^2 + 2(x + 3)(x^2 - 2)^3$$

$$y' = 2(x^2 - 2)^2(x + 3)[3x(x + 3) + x^2 - 2]$$

$$y' = 2(x^2 - 2)^2(x + 3)(4x^2 + 9x - 2)$$

① Mark for correct factorisation.

(c) $y = \sqrt[4]{4x + 3}$

2

$$y = (4x + 3)^{\frac{1}{4}}$$

$$y' = \frac{1}{4}(4x + 3)^{-\frac{3}{4}} \times 4$$

$$y' = (4x + 3)^{-\frac{3}{4}}$$

$$y' = \frac{1}{\sqrt[4]{(4x + 3)^3}}$$

① Mark for converting to fractional index

① Mark for correct solution

(d) $y = \frac{2x}{5 - x}$

2

$$u = 2x \quad v = 5 - x$$

$$u' = 2 \quad v' = -1$$

$$y' = \frac{2(5 - x) + 2x}{(5 - x)^2}$$

$$y' = \frac{10}{(5 - x)^2}$$

① Mark for recognising quotient rule

① Mark for correct solution

Question 15 (3 marks)

3

Find the equation of the normal to the curve $y = 2x^2 - 5x + 1$ at the point $(2, -1)$

$$\begin{array}{l|l}
 y' = 4x - 5 & y - y_1 = m(x - x_1) \\
 m = 4(2) - 5 & \\
 m_1 = 3 & y + 1 = -\frac{1}{3}(x - 2) \\
 m_2 = -\frac{1}{3} \quad (2, -1) & y + 1 = -\frac{1}{3}x + \frac{2}{3} \\
 & y = -\frac{1}{3}x - \frac{1}{3}
 \end{array}$$

Question 16 (2 marks)

2

Differentiate $f(x) = 2x + 3$ from first principles to show that $f'(x) = 2$.Hint: Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

$$\begin{array}{l|l}
 f(x) = 2x + 3 & f'(x) = \lim_{h \rightarrow 0} \frac{2x + 2h + 3 - 2x - 3}{h} \\
 f(x+h) = 2(x+h) + 3 & = \lim_{h \rightarrow 0} \frac{2h}{h} \\
 = 2x + 2h + 3 & = 2
 \end{array}$$

① Mark for differentiating

① Mark for finding gradient of normal

① Mark for correct equation.

① Mark for correctly substituting $f(x)$ and $f(x+h)$ into formula.

① Mark for correct simplification

Question 17 (2 marks)

2

Let $f(x) = 2x$ and $g(x) = \frac{x}{x+1}$. Solve $f(g(x)) = 3$.

$$f\left(\frac{x}{x+1}\right) = 2\left(\frac{x}{x+1}\right) = 3$$

$$\frac{2x}{x+1} = 3$$

$$2x = 3x + 3$$

$$-3 = x$$

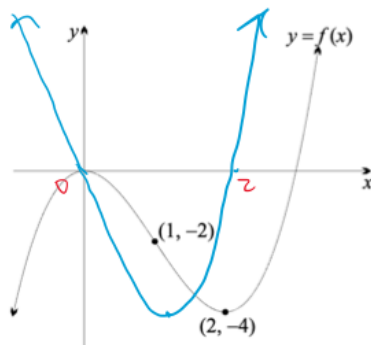
$$\therefore x = -3$$

Question 18 (2 marks)

2

The function $y = f(x)$ is graphed in the diagram below. The points $(2, -4)$ and the origin are stationary points and the point $(1, -2)$ is a point of inflexion.

Sketch the graph of the curve $y = f'(x)$ on the same cartesian plane, showing the shape of the curve and any x intercepts.



① Mark for correct substitution into correct function.

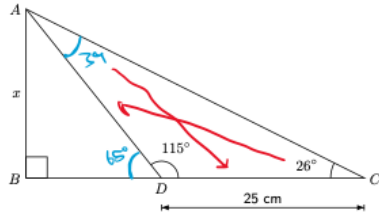
① Mark for correct value of x

① Mark for correct intercepts.

① Mark for correct shape.

Question 19 (3 marks)

3

Find the value of x , correct to 2 decimal places.

$$\angle BDA = 180 - 115 = 65$$

$$\angle DAC = 180 - 115 - 26 = 39$$

$$\frac{AD}{\sin 26} = \frac{25}{\sin 39}$$

$$AD = \frac{25 \sin 26}{\sin 39}$$

$$AD = 17.41$$

$$\therefore \sin 65 = \frac{x}{17.41}$$

$$x = 15.78 \text{ cm}$$

① Mark for identifying $\angle BDA = 65^\circ$

① Mark for correct use of sine rule

① Mark for correct use of trigonometry.

Question 20 (4 marks)

Sketch the graphs of the following, showing any asymptotes and any intercepts with the axes.

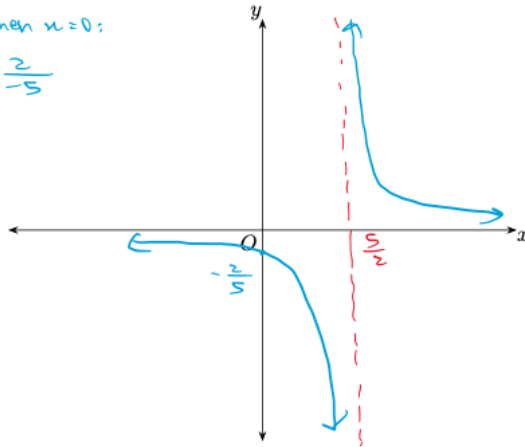
(a) $y = \frac{2}{2x-5}$

2

vertical asymptote: $2x-5 \neq 0$
 $2x \neq 5$
 $x \neq \frac{5}{2}$

y-int when $x=0$:

$$y = \frac{2}{-5}$$



① mark for calculating correct vertical asymptote and y-intercept.

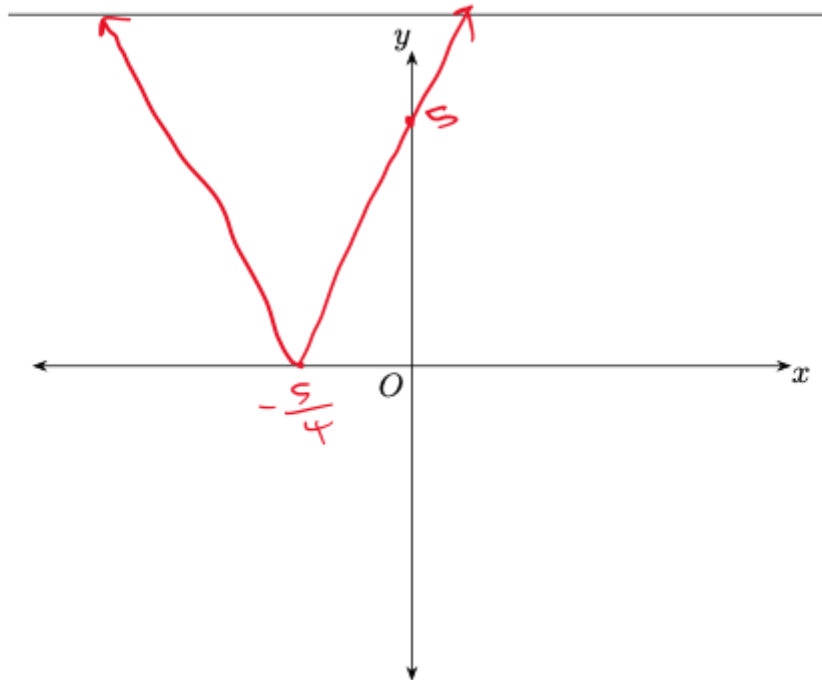
① mark for correct general shape of curve

(b) $y = |4x + 5|$

2

 y -int when $x = 0$

$y\text{-int} = 5$



Question 21 (3 marks)

3

Prove the identity $(\sec \theta + \tan \theta)^2 = \frac{1 + \sin \theta}{1 - \sin \theta}$.

L.H.S.

$$(\sec \theta + \tan \theta)^2$$

$$= \sec^2 \theta + 2 \sec \theta \tan \theta + \tan^2 \theta$$

$$= \frac{1}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} + 2 \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \sin^2 \theta}{\cos^2 \theta} + \frac{2 \sin \theta}{\cos^2 \theta}$$

$$= \frac{\sin^2 \theta + 2 \sin \theta + 1}{1 - \sin^2 \theta}$$

$$= \frac{(\sin \theta + 1)^2}{1 - \sin^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2}{(1 - \sin \theta)(1 + \sin \theta)}$$

$$= \frac{1 + \sin \theta}{1 - \sin \theta} = \text{R.H.S.}$$

① Mark for correct expansion.

① Mark for combining fractions together

① Mark for factorising difference of two squares.

Question 22 (3 marks)Consider the function $x^4 - 12x^2 + 32 = 0$.

- (a) Solve for
- x
- by first letting
- $u = x^2$
- .

$$u^2 - 12u + 32 = 0$$

$$(u - 8)(u - 4) = 0$$

$$u = 8$$

$$u = 4$$

$$x^2 = 8$$

$$x^2 = 4$$

$$x = \pm 2\sqrt{2}$$

$$x = \pm 2$$

- (b) If
- $x = \sin \theta$
- , explain why there are no solutions for
- θ
- .

lies outside amplitude for
the sine wave which is
between 1 and -1.

2 ① Mark for correct factorisation.

① Mark for finding correct solutions.

1 ① Mark for correct solutions.

Question 23 (8 marks)Consider the curve $f(x) = x^3 - 6x^2 + 9x - 4$.

(a) Find any stationary points and determine the nature.

3

$f'(x) = 3x^2 - 12x + 9$	$f''(x) = 6x - 12$
let $f'(x) = 0$ to find	sub $x = 1$ for concavity:
st. p + c	$f''(1) = 6 - 12$
$0 = 3(x^2 - 4x + 3)$	$= -6$
$0 = (x-3)(x-1)$	$\therefore$ concave down
$x = 1 \quad x = 3$	$f''(3) = 6(3) - 12$
$y = 0 \quad y = -4$	$= 6$
	$\therefore$ concave up

(b) Find the co-ordinates of the point of inflexion.

2

let $f''(x) = 0$
$6x - 12 = 0$
$x = 2 \quad y = -2$

① Mark for correct differentiation.

① Mark for finding stationary points

① Mark for proving concavity

① Mark for finding x-coordinate

①

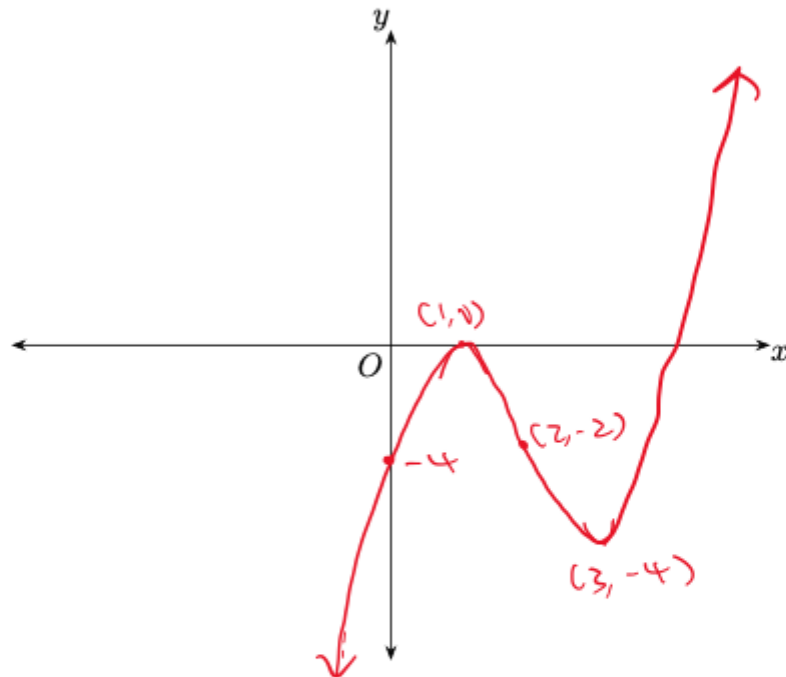
Must show change in concavity for second mark

by showing value not just sign

- (c) Hence, sketch the curve, showing all turning points, the y -intercept and the point of inflexion. 3

y -int: when $x = 0$

y -int = -4



Question 24 (4 marks)

4

Solve $2\cos^2 x + 3\sin x = 0$ for $0 \leq x \leq 2\pi$.

$$2(1 - \sin^2 x) + 3\sin x = 0$$

$$2 - 2\sin^2 x + 3\sin x = 0$$

$$0 = 2\sin^2 x - 3\sin x - 2$$

$$0 = 2\sin^2 x - 4\sin x + \sin x - 2$$

$$0 = 2\sin x(\sin x - 2) + (\sin x - 2)$$

$$0 = (\sin x - 2)(2\sin x + 1)$$

$$\sin x = 2 \quad \sin x = -\frac{1}{2}$$

↓

no sol

↓

$$\sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}$$

$$\therefore x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

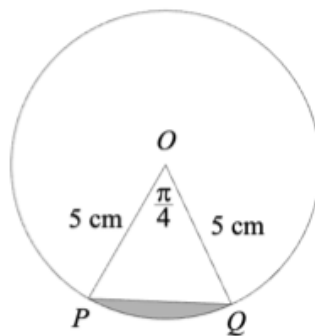
$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

① Mark for substitution of $1 - \sin^2 x$

① Mark for correct factorisation.

① Mark for finding $\sin x = 2$ and $\sin x = -\frac{1}{2}$

① Mark for finding all correct solutions

Question 25 (3 marks)

The diagram above shows a circle with centre O , radius 5 cm and $\angle POQ = \frac{\pi}{4}$. The points P and Q lie on the circumference of the circle.

- (a) Find the exact area of the sector OPQ .

1

$$A = \frac{1}{2} r^2 \theta$$

$$A = \frac{1}{2} \times 25 \times \frac{\pi}{4}$$

$$A = \frac{25\pi}{8} \text{ cm}^2$$

① Mark for correct answer

- (b) Find the exact area of $\triangle OPQ$

1

$$A = \frac{1}{2} ab \sin C$$

$$A = \frac{1}{2} \times 25 \times \sin \frac{\pi}{4}$$

$$A = \frac{25\sqrt{2}}{4} \text{ cm}^2$$

① Mark for correct answer

- (c) Hence find the exact area of the minor shaded segment.

1

$$A = \frac{25\pi}{8} - \frac{25\sqrt{2}}{4}$$

$$A = \frac{25\pi - 50\sqrt{2}}{8}$$

$$A = \frac{25(\pi - 2\sqrt{2})}{8} \text{ cm}^2$$

① Mark for correct answer,

Question 26 (5 marks)

Farmer Anneka wants to make a rectangular paddock with an area of 4000m^2 .

However, fencing costs are sky-high so she wants the paddock to have a minimum perimeter.

- (a) Show that the perimeter is given by the equation $P = 2x + \frac{8000}{x}$.

2

$$\begin{aligned} A &= x \times y & P &= 2x + 2y \\ 4000 &= x \times y & & \downarrow \\ & & P &= 2x + 2\left(\frac{4000}{x}\right) \\ y &= \frac{4000}{x} & P &= 2x + \frac{8000}{x} \end{aligned}$$

① Mark for making y the subject.
① Mark for correct substitution

- (b) Find the dimensions of the rectangle that will give the minimum perimeter. Give your answer correct to 1 decimal place.

3

$$\text{Let } P' = 0 \text{ For st. pt.}$$

① Mark for st. pt.

$$P = 2x + 8000x^{-1}$$

② Mark for finding minimum

$$P' = 2 - 8000x^{-2}$$

$$P' = 2 - \frac{8000}{x^2}$$

③ Mark for finding dimensions of rectangle

$$\text{Let } P' = 0$$

$$\frac{8000}{x^2} = 2$$

$$8000 = 2x^2$$

KAMBALA

CAUTION: No marks deducted for not explaining why positive v positive and negative. Always communicate mathematically why you chose positive. Otherwise, full marks will not be awarded in the future.

$$\begin{aligned} x^2 &= 4000 \\ x &= \pm \sqrt{4000} = \sqrt{4000} \quad (x > 0) \end{aligned}$$

$$P'' = 16000x^{-3}$$

$$P'' = \frac{16000}{x^3}$$

$$\text{sub } x = \sqrt{4000}$$

$$= \frac{16000}{(\sqrt{4000})^3}$$

$$= 0.06 \dots$$

= minimum

Max 3 marks awarded if the stationary point was not proven to be a minimum via a table of values or 2nd derivative test.

$\therefore$ at $x = \sqrt{4000}$ it will give minimum

$$A = x \times y$$

$$4000 = \sqrt{4000} \times y$$

$$y = \frac{4000}{\sqrt{4000}} \times \frac{\sqrt{4000}}{\sqrt{4000}}$$

$$= \sqrt{4000}$$

End of Task